ZCAM LATAM HUB / SYSTEM ADMINISTRATOR

Azure Databricks Platform - Target Architecture

Hub-and-Spoke | Production and Development isolation | Private data access | Disaster recovery

Controlled API egress

CORPORATE USERS

São Paulo office

Existing VPN connection

Entra ID / conditional access

EXISTING HUB VNET

Shared network services · CIDR to be confirmed

VPN GATEWAY

Existing corporate entry

São Paulo office

AZURE FIREWALL

FQDN-based HTTPS egress

Approved API destinations

PRIVATE DNS

Resolver + zone links

blob / dfs / sql / kv / adb

PRIVATE ACCESS

Workspace UI / SSO

Private endpoints + auth

EXTERNAL APIs × 2

HTTPS 443 only

FQDN allowlisting

Non-PII ingestion

Hub / Spoke peering

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PRODUCTION SUBSCRIPTION / Spoke VNet 10.20.0.0/16

AZURE DATABRICKS PREMIUM

VNet-injected classic compute · Secure Cluster Connectivity

Host + container subnets

Two delegated /24 subnets + NSGs

No public IP on compute

Private endpoints in PROD

ADB back-end · ADLS blob + dfs

SQL · Key Vault · Workspace access

DATA SERVICES / PUBLIC NETWORK ACCESS DISABLED

stlatamraw

Raw ingestion / non-PII

stlatamcurated

Customer PII

stlatamanalytics

Aggregated BI-ready

PRODUCTION KEY VAULT

RBAC · Private Link · secret rotation

UNITY CATALOG + ENTRA ID

Least privilege · auditing · PII grants

DEVELOPMENT SUBSCRIPTION / Spoke VNet 10.30.0.0/16

AZURE DATABRICKS PREMIUM

VNet-injected classic compute · Dev policies

Host + container subnets

Two delegated /24 subnets + NSGs

No public IP on compute

Private endpoints in DEV

ADB back-end · ADLS blob + dfs

Key Vault · Workspace access

DEV DATA SERVICES / ISOLATED

Raw / Curated / Analytics storage

Synthetic or irreversibly de-identified data only

DEVELOPMENT KEY VAULT

RBAC · Private Link · independent secrets

DEV ENTRA GROUPS + JOB POLICY

Auto-termination · no production PII

SQL over Private Link

DR replication / recovery

EXISTING AZURE SQL DATABASE

Hosted in another subscription · Production Private Endpoint

Entra authentication; DNS private resolution

Recovery and permissions require SQL owner approval

PRODUCTION DISASTER RECOVERY

Secondary approved region · Warm private networking foundation

Data replication + redeployable workspace and jobs

RPO target: 4h | RTO target: 2h | Validate with recovery drills

Legend: Blue = private network • Green = protected data / DR • Gold = restricted HTTPS egress

Proposed logical design • Not deployed • CIDRs / costs / RTO-RPO require validation